

TCS NQT Array and String Practice Questions

1. Find the Second Largest Number

A company stores the sales records of its employees in an array. Your task is to identify the employee with the second-highest sales figure.

Input:

5
10 20 30 40 50

Output:

40

2. Move All Zeros to End

A data processing system stores valid entries as non-zero values and invalid entries as zeros. Rearrange the array such that all invalid entries appear at the end while maintaining the order of valid entries.

Input:

6
1 0 2 0 4 5

Output:

1 2 4 5 0 0

3. Reverse a String

A company encrypts its employee IDs by reversing them. Given a string, write a program to generate its encrypted form.

Input:

TCSNQT

Output:

TQNSCT

4. Check Palindrome

A security system accepts only special access codes that remain the same when read from left to right and right to left.

Input:

MADAM

Output:

Palindrome

5. Find Missing Number

A warehouse stores product IDs numbered from 1 to N. Due to a system error, one product ID is missing from the database.

Input:

5

1 2 4 5

Output:

3

6. Character Frequency

A text analyzer needs to determine how many times a specific character appears in a given string.

Input:

TCSNQT

T

Output:

2

7. Remove Duplicates from Array

An online registration portal accidentally stores duplicate registration numbers. Remove duplicate entries and display unique registration numbers.

Input:

7

1 2 3 2 4 1 5

Output:

1 2 3 4 5

8. Check Anagram

Two words are considered equivalent if they contain the same characters with the same frequency.

Input:

listen

silent

Output:

Anagram

9. Rotate Array by K Positions

A circular conveyor belt contains N items. After every cycle, the belt rotates K positions to the right.

Input:

5

1 2 3 4 5

2

Output:

4 5 1 2 3

10. Longest Word in a Sentence

A text editor wants to highlight the longest word in a sentence for readability analysis.

Input:

TCS coding practice is important

Output:

important

11. Maximum Subarray Sum

A stock analyst records daily profit and loss values. Find the maximum profit that can be obtained from a continuous sequence of days.

Input:

8

-2 -3 4 -1 -2 1 5 -3

Output:

7

12. First Non-Repeating Character

A communication system receives a message stream. Find the first character that appears only once.

Input:

swiss

Output:

w

13. Largest and Smallest Element

An examination system stores marks of students. Find the highest and lowest marks obtained.

Input:

5

45 90 67 23 88

Output:

Largest = 90

Smallest = 23

14. Count Vowels and Consonants

A language analyzer needs to count vowels and consonants present in a word.

Input:

education

Output:

Vowels = 5

Consonants = 4

15. Longest Substring Without Repeating Characters

A network monitoring system records a stream of unique packet IDs. Find the length of the longest sequence without repetition.

Input:

abcabcbb

Output:

3